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Supplementary appendix 2

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Site and investigator list of FURLONG study

Site	Investigator(s)	Enrolment
National Cancer Center and National Clinical Research Center for Cancer and Cancer Hospital, Chinese Academy of Medical Sciences & Peking Union Medical College	Yuankai Shi	0
Harbin Medical University Cancer Hospital	Gongyan Chen	20
Hunan Cancer Hospital	Lin Wu / Nong Yang	20 (Lin Wu 12, Nong Yang 8)
Xuzhou Central Hospital	Xiang Wang	13
The First Hospital of China Medical University	Yunpeng Liu	13
The People's Hospital of Guangxi Zhuang Autonomous Region	Yanrong Hao	12
Cancer Hospital of Xinjiang Medical University	Chunling Liu	11
The Affiliated Hospital of Xuzhou Medical College	Shuyang Zhu	11
Nantong Cancer Hospital	Xiaodong Zhang	11
The First Affiliated Hospital of Wenzhou Medical University	Yuping Li	11
The First Affiliated Hospital of Dalian Medical University	Jiwei Liu	10
The First Affiliated Hospital of USTC	Lejie Cao	10
Jilin Cancer Hospital	Ying Cheng	10
The Second Hospital of Anhui University	Hui Zhao	9
Beijing Chest Hospital	Shucai Zhang	9

Affiliated Hospital of Hebei University	Aimin Zang	8
The First Hospital of Jilin University	Jiuwei Cui	8
Affiliated Hospital of Nantong University	Jian Feng	8
Tianjin Medical University Cancer Institute & Hospital	Changli Wang / Richeng Jiang	7
The First Affiliated Hospital of Xinxiang Medical University	Hongrui Niu	7
Shanxi Province People’s Hospital	Mei He	7
The Second People’s Hospital of Neijiang	Ou Jiang	7
General Hospital of Ningxia Medical University	Xiuzhong Li	7
Henan Cancer Hospital	Zhiyong Ma	7
Affiliated Cancer Hospital and Institute of Guangzhou Medical University	Chuan Jin	6
The First Affiliated Hospital of Xiamen University	Feng Ye	6
Guangxi Medical University Affiliated Tumor Hospital	Qitao Yu	6
Liaoning Cancer Hospital & Institute	Xiaoling Li	6
Taihe Hospital	Zhiguo Luo	6
The First Affiliated Hospital of Guangzhou Medical University	Chengzhi Zhou	5
Army Medical Center of PLA	Dong Wang	5
Weifang People's Hospital	Guohua Yu	5
Zhongshan Hospital	Qunying Hong	5
Henan Province People's Hospital	Shundong Cang	5
Nanjing First Hospital	Wei Gu	5
The Affiliated Hospital of Inner Mongolia Medical University	Xiuhua Fu	5

Lishui City People's Hospital	Zhuo Cao	5
The Second Affiliated Hospital of Nanchang University	Anwen Liu	4
Liuzhou People’s Hospital	Jingzhang Li	4
The First Affiliated Hospital, Zhejiang University	Jianying Zhou	4
Shanghai Chest Hospital	Liyan Jiang	4
Union Hospital, Tongji Medical College, Huazhong University of Science and Technology	Li Liu	4
The First People's Hospital of Changzhou	Mei Ji	4
The First Affiliated Hospital of Zhengzhou University	Qingxia Fan	4
PLA Cancer Centre of Nanjing Bayi Hospital	Shukui Qin	4
Zhejiang Cancer Hospital	Yiping Zhang	4
Beijing Cancer Hospital	Ziping Wang	4
Beijing Cancer Hospital	Jian Fang	3
Affiliated Hospital of Guilin Medical University	Libing Ma	3
The Second Affiliated Hospital Zhejiang University School of Medicine	Jianjin Huang	2
Sir Run Run Shaw Hospital Zhejiang University School of Medicine	Kejing Ying	2
People's Hospital of Xinjiang Uygur Autonomous Region	Jiang Liu	1
Peking Union Medical College Hospital	Mengzhao Wang	1
Shanghai Changzheng Hospital	Bing Li	0
The Second Hospital of Dalian Medical	Yang Zhang	0

University		

Protocol amendment and interim analysis.

In this study, we used a group sequential design with one prespecified interim analysis to evaluate the efficacy and safety of the treatment. The first version of clinical study protocol dated November 25, 2018 included a sample size of 358 and progress free survival (PFS) events of 192. A protocol amendment dated November 15, 2019 was made to add one blinded efficacy data evaluation for the purpose of the re-assessment of sample size. The blinded data was evaluated by the Sponsor Committee (consisting of company internal medical and statistical experts) after approximately 64 PFS events (1/3 of 192 events) have been observed. Based on the results of recently reported external data and the primary efficacy results of the pooled data in current study, a decision of the sample size adjustment was made by the Sponsor Committee. With updating statistical assumption of HR from 0.647 to 0.688, the study needs approximately 238 events of progression or death in a total of 358 randomly assigned patients to provide at least 80% power detecting a hazard ratio of 0.688 using a log-rank test at an overall two-sided alpha level of 5%.

An interim analysis was planned after the observation of approximately 119 events (50% of planned events for the final analysis) in this study. The O'Brien-Fleming stopping boundary at the time of interim analysis for the primary endpoint was calculated with the use of a Lan-DeMets alpha spending function. A two-sided alpha level of 0.00306 and 0.049 were prespecified for the primary endpoint at interim analysis and final analysis respectively. The actual alpha level was calculated on the basis of the number of events observed at the interim analysis. The interim analysis was conducted by an independent data monitoring committee (IDMC). If the results of the primary endpoint were statistically significant at the interim analysis based on prespecified alpha spending, IDMC may recommend stopping the study for efficacy. By the cutoff

date of July 1st, 2020, 117 events (49.2% of planned events of final analysis) were observed. The interim analysis was conducted by IDMC. Based on the results of the interim analysis, the IDMC recommended that the trial be continued without modification.

Supplementary Table 1. Major protocol violations of patients excluded from per-protocol set.

Patient NO.	Group	Major protocol violations
1	Furmonertinib	This patient was found having received osimertinib for 3 months before enrollment and met the exclusion criteria.
2	Furmonertinib	This patient had brain and nasopharyngeal lesions at screening recorded as non-target lesions. After treatment, all lesions improved except for the nasopharyngeal lesion, and was found to be nasopharyngeal carcinoma. Thus, one of the exclusion criteria was met.
3	Gefitinib	The pathological report of the target lung lesion of this patient was adenocarcinoma, based on which this patient was enrolled. This patient also had a lesion on the frontal bone at screening. After treatment, the target lesion in the lung was stable but the frontal bone lesion progressed gradually and then SCLC cells were found in this lesion. Based on the investigator's judgement, the frontal bone lesion might be a metastatic site from a mixed pathological type lung cancer. Thus, this patient met the exclusion criteria.

Supplementary Table 2. Summary of response in the full analysis set assessed by IRC

	Furmonertinib (n=178)	Gefitinib (n=179)	p value*
Response			
Complete response	0	0	..
Partial response	158 (89%)	151 (84%)	..
Stable disease	12 (7%)	16 (9%)	..
Progressed disease	5 (3%)	10 (6%)	..
Not evaluable	1 (1%)	1 (1%)	..
Not evaluated	2 (1%)	1 (1%)	..
Objective response	89% (95%CI 83-93)	84% (95%CI 78-89)	0·2078
Disease control	96% (95%CI 91-98)	93% (95%CI 89-97)	0·3551

Data are n (%) or % (95%CI). IRC=independent review center. *Calculated with Wald χ^2 test.

Supplementary Table 3. Summary of investigator assessed endpoints in the full analysis set

Endpoints	Furmonertinib (n=178) versus Gefitinib (n=179)
Progression-free survival, month (95%CI)	18.3 (15.2-20.8) vs 11.0 (9.7-12.4)
HR (95%CI), p	HR 0.51 (0.39-0.66), p<0.0001
Objective response rate, (95%CI)	75% (78-81) vs 69% (61-75)
OR (95%CI), p	OR 1.38 (0.85-2.22), p=0.1905
Disease control rate, (95%CI)	95% (91-98) vs 89% (84-94)
OR (95%CI), p	OR 2.31 (1.00-5.36), p=0.0502
Duration of response, month (95%CI)	18.0 (16.5-22.0) vs 9.7 (8.3-11.8)
HR (95%CI), p	HR 0.43 (0.31-0.59), p<0.0001
Depth of response, median (range)	-47% (-100, 51) vs -46% (-100, 53)
Mean difference (95%CI), p	Mean difference -1 (-6, 4), p=0.5673
Time to progression, month (95%CI)	20.3 (15.5-20.9) vs 11.0 (9.7-20.4)
HR (95%CI), p	HR 0.48 (0.37-0.62), p<0.0001

HR=hazard ratio; OR=odds ratio.

Supplementary Table 4. Summary of adverse events

	Furmonertinib (n=178)	Gefitinib (n=179)
Treatment-emergent adverse events		
Any adverse event	178 (100%)	177 (99%)
Grade ≥3 adverse event	62 (35%)	60 (34%)
Any serious adverse event	44 (25%)	29 (16%)
Dose interruption due to adverse event	27 (15%)	34 (19%)
Dose reduction due to adverse event	5 (3%)	6 (3%)
Discontinuation due to adverse event	11 (6%)	7 (4%)
Any adverse event with outcome of death	10 (6%)	3 (2%)
Treatment-related adverse events†		
Any adverse event	160 (90%)	170 (95%)
Grade ≥3 adverse event	20 (11%)	32 (18%)
Any serious adverse event	10 (6%)	11 (6%)
Dose interruption due to adverse event	24 (13%)	28 (16%)
Dose reduction due to adverse event	5 (3%)	NA*
Discontinuation due to adverse event	6 (3%)	4 (2%)
Any adverse event with outcome of death	0	0

Data are n (%). NA=not available. * Dose reduction of gefitinib was not allowed in the protocol so only furmonertinib placebo was reduced. †Assessed by investigators.

Supplementary Table 5. Treatment-related serious adverse events

	Furmonertinib (n=178)			Gefitinib (n=179)		
	Grade 1-2	Grade 3-4	Grade 5	Grade 1-2	Grade 3-4	Grade 5
Elevated alanine aminotransferase	2 (1%)	2 (1%)	0	1 (1%)	2 (1%)	0
Elevated aspartate	2 (1%)	1 (1%)	0	1 (1%)	3 (2%)	0

aminotransferase						
Cerebral infarction	1 (1%)	1 (1%)	0	0	0	0
Gastritis	1 (1%)	0	0	0	0	0
Gastroenteritis	1 (1%)	0	0	0	0	0
Abnormal liver function	0	1 (1%)	0	0	1 (1%)	0
Decreased fibrinogen	0	1 (1%)	0	0	0	0
Decreased platelet count	0	1 (1%)	0	0	0	0
Diarrhea	0	1 (1%)	0	0	0	0
Obstructive pancreatitis	0	1 (1%)	0	0	0	0
Cholecystitis	0	1 (1%)	0	0	0	0
Cholelithiasis	0	1 (1%)	0	0	0	0
Non-infectious cystitis	0	0	0	1 (1%)	0	0
Weight loss	0	0	0	0	1 (1%)	0
Intestinal obstruction	0	0	0	0	1 (1%)	0
Bleeding from gastric ulcer	0	0	0	0	1 (1%)	0
Furuncle	0	0	0	0	1 (1%)	0
Chest discomfort	0	0	0	0	1 (1%)	0

Data are n (%).

Supplementary Table 6. Adverse events with an outcome of death*

	Furmonertinib (n=178)	Gefitinib (n=179)
Respiratory failure	2 (1%)	0
Death of unconfirmed reason	1 (1%)	2 (1%)
Pulmonary embolism	1 (1%)	0
Fall	1 (1%)	0
Chemical poisoning of glyphosate	1 (1%)	0
Malnutrition	1 (1%)	0
Cerebral infarction	1 (1%)	0
Paraneoplastic syndrome	1 (1%)	0
Venous embolism	1 (1%)	0
Multiple organ dysfunction syndrome	0	1 (1%)

Data are n (%). *All of these events were assessed by the investigators as possibly not-related to study treatment.

Supplementary Table 7. Overall changes of adjusted least square mean of QLQ-C30 and QLQ-LC13 scores from baseline.

QLQ-C30 scores, least square mean changes from baseline (95%CI)				
	Furmonertinib N=178	Gefitinib N=179	Mean Difference	p-value
Global health status	3.80 (1.76, 5.85)	2.60 (0.51, 4.69)	1.21 (-1.71, 4.13)	0.4165
Functional scales				
Physical functioning	3.90 (2.58, 5.22)	1.76 (0.40, 3.11)	2.14 (0.25, 4.04)	0.0267
Role functioning	1.86 (0.25, 3.47)	0.58 (-1.07, 2.22)	1.29 (-1.01, 3.59)	0.2719
Emotional functioning	3.17 (1.89, 4.45)	3.05 (1.74, 4.35)	0.12 (-1.7, 4.20)	0.8932
Cognitive functioning	0.20 (-1.14, 1.55)	-0.96 (-2.35, 0.43)	1.16 (-0.77. 3.10)	0.2375
Social functioning	1.69 (-0.38, 3.77)	0.16 (-1.96, 2.29)	1.53 (-1.44, 4.50)	0.3110
Symptom scales/items				
Fatigue	-7.22 (-8.87, -5.56)	-6.00 (-7.71, -4.30)	-1.21 (-3.59, 1.16)	0.3165
Nausea and vomiting	-2.29 (-3.03, -1.55)	-0.73 (-1.50, 0.03)	-1.56 (-2.62, -0.49)	0.0043
Pain	-8.17 (-9.86, -6.66)	-6.12 (-7.70, -4.54)	-2.05 (-4.24, 0.14)	0.0663
Dyspnea	-6.47 (-8.21, -4.73)	-4.83 (-6.63, -3.03)	-1.64 (-4.14, 0.87)	0.1988
Insomnia	-7.22 (-8.80, -5.64)	-6.72 (-8.36, -5.08)	-0.50 (-2.78, 1.78)	0.6662
Appetite loss	-4.26 (-5.56, -2.86)	-2.02 (-3.47, -0.56)	-2.24 (-4.26, -0.23)	0.0293
Constipation	-1.58 (-2.74, -0.42)	-1.98 (-3.19, -0.77)	0.40 (-1.28, 2.07)	0.6422
Diarrhea	1.93 (0.67, 3.19)	5.29 (3.97, 6.61)	-3.36 (-5.19, -1.54)	0.0003
Financial difficulties	-7.27 (-10.04, -4.49)	-5.19 (-8.03, -2.35)	-2.08 (-6.05, 1.89)	0.3041
QLQ-L13 scores				
Dyspnea	-6.65 (-8.27, -5.04)	-4.51 (-6.16, -2.85)	-2.15 (-4.46, 0.16)	0.0683
Cough	-15.08 (-17.21, -12.96)	-16.28 (-18.47, -14.09)	1.20 (-1.85, 4.25)	0.4403
Haemoptysis	-1.72	-1.52	-0.20	0.6627

	(-2.36, -1.08)	(-2.17, -0.87)	(-1.12, 0.71)	
Sore mouth	1.62 (0.28, 2.95)	2.56 (1.17, 3.94)	-0.94 (-2.86, 0.98)	0.3358
Dysphagia	-0.95 (-2.05, 0.14)	0.00 (-1.14, 1.13)	-0.95 (-2.53, 0.63)	0.2377
Peripheral neuropathy	-2.48 (-3.78, -1.18)	-2.79 (-4.15, -1.43)	0.31 (-1.58, 2.19)	0.7499
Alopecia	-0.51 (-1.83, 0.82)	2.12 (0.73, 3.50)	-2.62 (-4.54, -0.71)	0.0074
Chest pain	-10.39 (-12.20, -8.57)	-8.86 (-10.75, -6.98)	-1.52 (-4.14, 1.10)	0.2533
Pain in arm or shoulder	-7.04 (-8.74, -5.34)	-5.71 (-7.47, -3.94)	-1.33 (-3.79, 1.12)	0.2863
Pain in other parts	-8.44 (-10.38, -6.49)	-3.88 (-5.91, -1.86)	-4.55 (-7.37, -1.74)	0.0016

Supplementary Table 8. Subsequent anti-tumor therapies post study drugs irrespective of treatment lines.

	Furmonertinib (n=178)	Gefitinib (n=179)
First-generation EGFR-TKIs	20 (11)	25 (14)
Gefitinib	9 (5)	19 (11)
Erlotinib	1 (1)	0
Icotinib	11 (6)	6 (3)
Second-generation EGFR-TKIs	5 (3)	4 (2)
Afatinib	4 (2)	4 (2)
Dacomitinib	1 (1)	0
Third-generation EGFR-TKIs	22 (12)	77 (43)
Osimertinib	16 (9)	56 (31)
Aumolertinib	8 (4)	20 (11)
Furmonertinib	1 (1)	2 (1)
ASK120067	0	1 (1)
SH-1028	0	3 (2)
Chemotherapy	52 (29)	57 (32)
Immunotherapy	19 (11)	12 (7)
Anti-PD-1	17 (10)	10 (6)
Anti-PD-L1	2 (1)	2 (1)
Antiangiogenesis therapy	38 (21)	46 (26)
Bevacizumab	25 (14)	35 (20)
Anlotinib	12 (7)	9 (5)
Others	5 (3)	6 (3)
Traditional Chinese medicine	8 (4)	22 (12)
Others	2 (1)	6 (3)

EGFR=epidermal growth factor receptor.

Supplementary Table 9. Efficacy in patients with concurrent *EGFR* Exon 20 insertion mutations

Group	Sex	Age, years	CNS metastases	*EGFR* sensitizing mutation	*EGFR* concurrent mutation	Progression-free survival, m
Furmonertinib	F	59	Y	L858R	Ex20ins	9.7
Furmonertinib	F	41	Y	L858R	Ex20ins	20.6
Gefitinib	F	70	N	L858R	Ex20ins	3.4
Gefitinib	F	45	N	L858R	Ex20ins	5.6

EGFR=epidermal growth factor receptor; CNS=central nervous system.

Supplementary Table 10. Treatment-emergent adverse events

	Furmonertinib (n=178)			Gefitinib (n=179)		
	Grade 1-2	Grade 3-4	Grade 5	Grade 1-2	Grade 3-4	Grade 5
Diarrhea	50 (28%)	4 (2%)	0	75 (42%)	0	0
Elevated alanine aminotransferase	50 (28%)	2 (1%)	0	98 (55%)	13 (7%)	0
Elevated aspartate aminotransferase	47 (26%)	2 (1%)	0	90 (50%)	11 (6%)	0
Rash	32 (18%)	0	0	71 (40%)	2 (1%)	0
Weight gain	31 (17%)	0	0	31 (17%)	0	0
Decreased white blood cell count	30 (17%)	0	0	24 (13%)	0	0
Anemia	29 (16%)	2 (1%)	0	20 (11%)	5 (3%)	0
Upper respiratory tract infection	28 (16%)	2 (1%)	0	30 (17%)	3 (2%)	0
Urinary tract infection	27 (15%)	1 (1%)	0	31 (17%)	0	0
Oral ulcer	25 (14%)	0	0	16 (9%)	0	0
Cough	24 (13%)	0	0	21 (12%)	0	0
Increased serum creatinine	22 (12%)	1 (1%)	0	6 (3%)	0	0
Constipation	22 (12%)	0	0	13 (7%)	0	0

Decreased neutrophil count	20 (11%)	1 (1%)	0	13 (7%)	0	0
Vomiting	19 (11%)	1 (1%)	0	16 (9%)	0	0
Weight loss	19 (11%)	0	0	18 (10%)	2 (1%)	0
Decreased platelet count	18 (10%)	2 (1%)	0	11 (6%)	0	0
Chest pain	18 (10%)	0	0	14 (8%)	0	0
Joint pain	17 (10%)	0	0	21 (12%)	0	0
Increased γ-glutamyltransferase	14 (8%)	3 (2%)	0	22 (12%)	2 (1%)	0
Decreased lymphocyte count	13 (7%)	1 (1%)	0	11 (6%)	0	0
Headache	13 (7%)	1 (1%)	0	10 (6%)	0	0
Nausea	13 (7%)	0	0	18 (10%)	1 (1%)	0
Back pain	12 (7%)	0	0	19 (11%)	0	0
ECG QT prolongation	10 (6%)	7 (4%)	0	10 (6%)	5 (3%)	0
Increased blood creatine phosphokinase	10 (6%)	1 (1%)	0	8 (4%)	1 (1%)	0
Pruritis	10 (6%)	0	0	18 (10%)	0	0
Stomatitis	10 (6%)	0	0	16 (9%)	1 (1%)	0
Dyspnea	8 (4%)	2 (1%)	0	5 (3%)	1 (1%)	0
Hypocalcemia	7 (4%)	0	0	6 (3%)	1 (1%)	0
Increased	6 (3%)	4 (2%)	0	10 (6%)	2 (1%)	0

blood pressure						
Hypokalemia	6 (3%)	3 (2%)	0	19 (11%)	2 (1%)	0
Increased amylase	6 (3%)	1 (1%)	0	8 (4%)	1 (1%)	0
Abdominal pain	6 (3%)	0	0	5 (3%)	1 (1%)	0
Epistaxis	5 (3%)	1 (1%)	0	8 (4%)	1 (1%)	0
Pleural effusion	5 (3%)	1 (1%)	0	6 (3%)	2 (1%)	0
Hypertension	4 (2%)	4 (2%)	0	4 (2%)	3 (2%)	0
Pneumonia	4 (2%)	4 (2%)	0	3 (2%)	3 (2%)	0
Chest discomfort	4 (2%)	0	0	1 (1%)	1 (1%)	0
Increased bilirubin	3 (2%)	1 (1%)	0	32 (18%)	1 (1%)	0
Cerebral infarction	2 (1%)	3 (2%)	1 (1%)	0	1 (1%)	0
Hyperkalemia	2 (1%)	1 (1%)	0	1 (1%)	2 (1%)	0
Protrusion of intervertebral disc	2 (1%)	1 (1%)	0	1 (1%)	0	0
Contrast agent allergy	2 (1%)	1 (1%)	0	0	1 (1%)	0
Cystitis	2 (1%)	1 (1%)	0	0	0	0
Pulmonary embolism	2 (1%)	0	1 (1%)	1 (1%)	0	0
Increased conjugated	2 (1%)	0	0	6 (3%)	1 (1%)	0

bilirubin						
Musculoskeletal pain in the chest	2 (1%)	0	0	3 (2%)	1 (1%)	0
Paronychia	1 (1%)	1 (1%)	0	13 (7%)	0	0
Gingivitis	1 (1%)	1 (1%)	0	1 (1%)	0	0
Pericardial effusion	1 (1%)	0	0	2 (1%)	1 (1%)	0
Asthma	1 (1%)	0	0	1 (1%)	2 (1%)	0
Cancer pain	1 (1%)	0	0	1 (1%)	1 (1%)	0
Bronchitis	1 (1%)	0	0	1 (1%)	1 (1%)	0
Decreased glomerular filtration rate	1 (1%)	0	0	0	1 (1%)	0
Abnormal liver function	0	1 (1%)	0	2 (1%)	2 (1%)	0
Herpes zoster	0	1 (1%)	0	1 (1%)	0	0
Atrial fibrillation	0	1 (1%)	0	1 (1%)	0	0
Cholelithiasis	0	1 (1%)	0	1 (1%)	0	0
Cholecystitis	0	1 (1%)	0	0	0	0
Decreased fibrinogen	0	1 (1%)	0	0	0	0
Abdominal infection	0	1 (1%)	0	0	0	0
Wound infection	0	1 (1%)	0	0	0	0
Gout	0	1 (1%)	0	0	0	0
Anasarca	0	1 (1%)	0	0	0	0

Syncope	0	1 (1%)	0	0	0	0
Vertebrobasilar insufficiency	0	1 (1%)	0	0	0	0
Sacroiliitis	0	1 (1%)	0	0	0	0
Thrombocythemia	0	1 (1%)	0	0	0	0
Type I hypersensitivity	0	1 (1%)	0	0	0	0
Obstructive pancreatitis	0	1 (1%)	0	0	0	0
Urinary retention	0	1 (1%)	0	0	0	0
Oliguria	0	1 (1%)	0	0	0	0
Femoral fractures	0	1 (1%)	0	0	0	0
Pancreatic cancer	0	1 (1%)	0	0	0	0
Respiratory failure	0	0	2 (1%)	0	0	0
Death	0	0	1 (1%)	0	0	2 (1%)
Venous embolism	0	0	1 (1%)	0	0	0
Malnutrition	0	0	1 (1%)	0	0	0
Fall	0	0	1 (1%)	0	0	0
Chemical poisoning of glyphosate	0	0	1 (1%)	0	0	0
Paraneoplasti	0	0	1 (1%)	0	0	0

c syndrome						
Arrhythmia	0	0	0	4 (2%)	1 (1%)	0
Furuncle	0	0	0	1 (1%)	1 (1%)	0
Trichiasis	0	0	0	1 (1%)	1 (1%)	0
Increased systolic pressure	0	0	0	0	1 (1%)	0
Increased hemoglobin	0	0	0	0	1 (1%)	0
Intestinal obstruction	0	0	0	0	1 (1%)	0
Bleeding from gastric ulcer	0	0	0	0	1 (1%)	0
Heart failure	0	0	0	0	1 (1%)	0
Bleeding	0	0	0	0	1 (1%)	0
Blood coagulation disorders	0	0	0	0	1 (1%)	0
Ulcerative keratitis	0	0	0	0	1 (1%)	0
Arulsion	0	0	0	0	1 (1%)	0
Mantle cell lymphoma	0	0	0	0	1 (1%)	0
Multiple organ dysfunction syndrome	0	0	0	0	0	1 (1%)

Data are n (%). Grade 1 or 2 treatment-related adverse events of at least 10% in either group and all grade 3–5 treatment-related adverse events are reported.

Supplementary Table 11. Detailed information of ECG QT prolongation

	Furmonertinib (n=178)	Gefitinib (n=179)
QTcF after treatment		
>480 ms, ⩽500 ms	6 (3%)	3 (2%)
>500 ms	4 (2%)	4 (2%)
QTcF change from baseline		
>60 ms, ⩽90 ms	13 (7%)	12 (7%)
>90 ms	3 (2%)	2 (1%)
QTcF >500 ms and prolongation >60 ms	4 (2%)	4 (2%)

Data are n (%).

Supplementary Table 12. Screening failure reasons.

Screening failure reasons*		Number of patients
Did not meet inclusion criteria	Inclusion criterion 1	1
	Inclusion criterion 3	3
	Inclusion criterion 4	6
	Inclusion criterion 5	233
	Inclusion criteria 5/6	7
	Inclusion criterion 6	21
	Inclusion criterion 8	11
Met exclusion criteria	Exclusion criterion 1	5
	Exclusion criterion 4	12
	Exclusion criterion 5	4
	Exclusion criterion 6	2
	Exclusion criterion 7	21
	Exclusion criterion 8	12
	Exclusion criterion 9	8
	Exclusion criterion 10	1
	Exclusion criteria 10/13	1
	Exclusion criterion 13	5
Withdrawal of consent		27
Decided by investigator		1
Other		21

*Screening was based on protocol version 2.0. For detailed inclusion/exclusion criteria, please refer to supplementary table 12.

Supplementary Table 13. Inclusion/exclusion criteria of protocol 2.0.

Criteria	No.	Detail
Inclusion criteria	1	Only after conforming to all of following criteria, the patients can be enrolled into this study; the subjects volunteer to participate in this study, and sign the informed consent form.
	2	Male or female subjects aged 18 above (18 inclusive).
	3	ECOG performance status score is 0-1; no worsening within 2 weeks prior to the scoring of ECOG performance status; subjects with life expectancy ≥ 12 weeks judged by the investigator
	4	Histopathologically diagnosed, unresectable and incurable, treatment-naïve locally advanced or metastatic pulmonary adenocarcinoma (e.g., can be systemic recurrence after previous operation for early tumor, or newly diagnosed stage ⅢB/Ⅳ tumor. AJCC version 8 is used as the staging system for lung cancer);
	5	Diagnosed as localized advanced or metastatic lung adenocarcinoma through the pathological examination of tissue sample, and verifed as at least one sensitive mutation of *EGFR* in Ex19del or L858R gene (either single or mixed mutation gene) through the test at center laboratory.
	6	Patients must provide adequate tumor tissue sections (can not be sourced from the tumor lesion locally treated, such as irradiation, the newly occurred lesion following local therapy is acceptable) in screening period, for detection of *EGFR* mutation status at central laboratory, please see Central Laboratory Manual for the details.
	7	No prior systemic anti-tumor therapy for advanced/metastatic non-small cell lung cancer. Patients who have received adjuvant therapy or neoadjuvant therapy (chemotherapy, radiotherapy or other therapy) can be enrolled if there is no progression within one year after treatment; for the patients who have received local therapy (radiotherapy or perfusion therapy in pleural cavity), they can participate in the study if the lesions in the range of local therapy are non-target lesions.
	8	At least one tumor lesion can meet the following requirements in accordance with the response evaluation criteria in solid tumors (RECIST 1.1): it is not irradiated previously or biopsied for the screening period (if there is only one target lesion that must receive tissue biopsy, baseline tumor evaluation needs to be performed at least

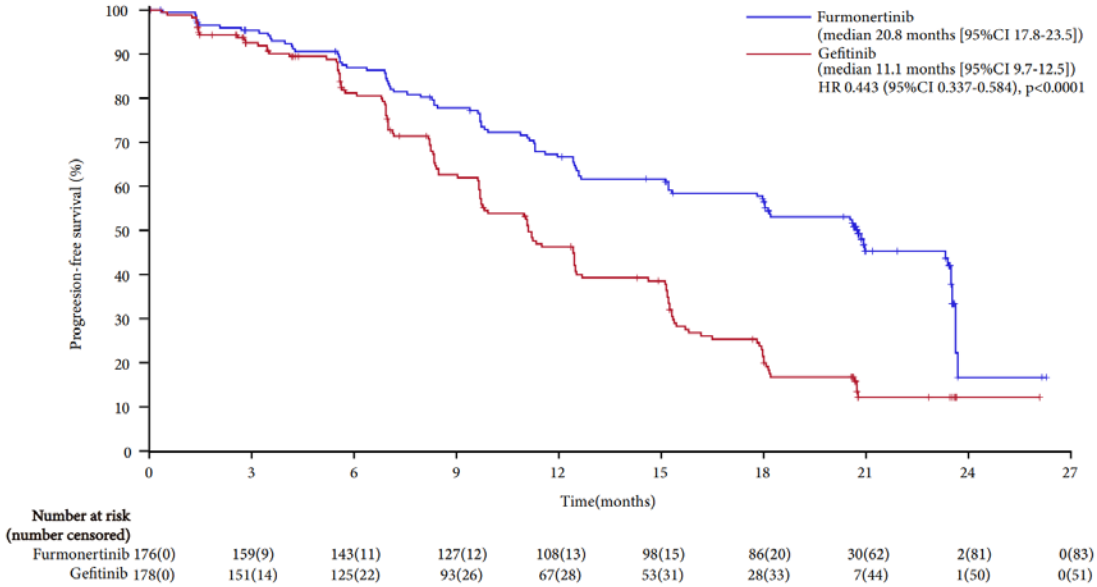
		14 days after the biopsy in screening period), and can be accurately measured at baseline, the longest diameter will be ≥10mm at baseline (short diameter is required to be ≥15mm for lymph nodes). The measurement method selected is suitable for repeated measurement accurately, and can be computed tomography (CT) or magnetic resonance imaging (MRI). If there is only one measurable lesion, it will be required not to be irradiated previously.
	9	Females of childbearing potential must have a negative serum or urine pregnancy test within 7 days prior to the first dose. Adequate contraceptive measures must be taken for male patients and female patients of childbearing potential from the signature of informed consent form to 3 months after the last dose of study drug. See 3.6 for details: Contraceptive measures, definition of Females of childbearing potential and contraceptive requirement.
Exclusion criteria	1	Having received any one of the following therapies prior to randomization: • Previously receiving systematic antitumor therapy for advanced/metastatic non-small cell lung cancer (e.g., standard chemotherapy, targeted therapy, biotherapy, immunotherapy, etc.); please refer to the inclusion criterion 7 for neoadjuvant and adjuvant therapies; • Previously receiving other systematic anticancer therapy for advanced/metastatic non-small cell lung cancer; • Patients who have received intrapleural perfusion therapy can be enrolled only after the hydrothorax is stable for 28 days or above; • Prior systemic anti-tumor therapy of EGFR-TKI; • Having received major operation within 28 days prior to the first dose of study drug (in China, for the definition of major surgical operation, please refer to grade 3 and 4 surgery specified in the Management Method for Clinical Application of Medical Technology implemented on May 1, 2009); • Having received radiation in an area of ⩾30% bone marrow or a wide range of radiotherapy within 28 days prior to the first dose of study drug; having received local radiotherapy or palliative radiotherapy for bone metastasis within 14 days prior to the first dose of study drug; • Having received CYP3A4 potent inhibitor or inducer within 7 days prior to the first dose, or need to continue treatment with these drugs during the study (drug list is

		provided in Appendix 6 in detail); • Having received the traditional Chinese medicine and Chinese patent drug preparation indicated for tumors within 7 days prior to the first dose, or need to continue treatment with these drugs during the study; • Currently receiving the drug therapy known to prolong QTc interval, and need to continue treatment with these drugs during the study (drug list is provided in Appendix 7 in detail); • Discontinuation of other clinical investigational products for less than 14 days prior to the first dose.
	2	The toxicity associated with previous anti-tumor treatment do not recover to ≤CTCAE grade 1, except for alopecia and ≤CTCAE grade 2 peripheral neurotoxicity caused by chemotherapy; please refer to the exclusion criteria 7 for bone marrow and hepatic and renal functions.
	3	Mixed histological type, i.e., pulmonary adenocarcinoma mixed with squamous cell carcinoma.
	4	With spinal cord compression or symptomatic brain metastasis (except symptomless state, stable disease, unnecessary for steroid treatment of ≥28 days before the starting of study drug); at symptom stabilization of ≥28 days after the completion of localized radiotherapy of brain metastasis.
	5	Having other malignant tumor or history of other malignant tumor; except basal cell carcinoma of skin, carcinoma in situ of cervix and ductal carcinoma in situ that have been effectively controlled.
	6	Clinically serious gastrointestinal dysfunction that may affect the intake, transport and absorption of study drug, for example, inability to take drugs orally, uncontrollable nausea or vomiting, history of large area gastrectomy, uncured recurrent diarrhea, atrophic gastritis (age < 60 years at the onset of disease), uncured gastric disorder requiring long-term use of PPI antacid, Crohn's disease, ulcerative colitis.
	7	Inadequate bone marrow reserve and hepatic and renal functions with the following laboratory values (no transfusion of blood or blood products, no use of granulocyte colony-stimulating factor or other hematopoietic stimulating factors for correction within two weeks prior to blood collection for laboratory test for enrollment): • Absolute neutrophil count < 1.5 × 109/L;

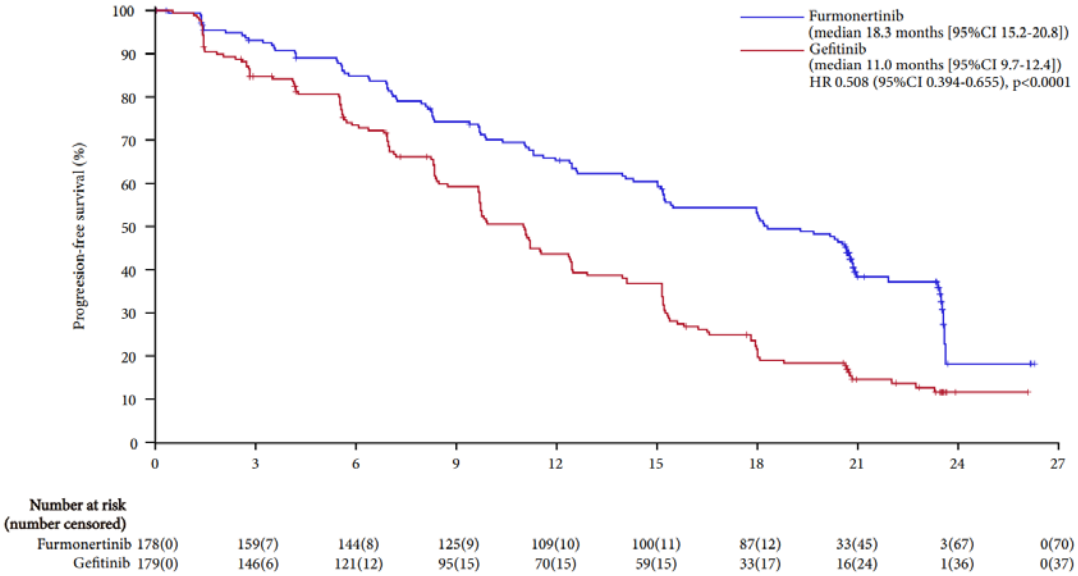
		• Haemoglobin <90g/L; • Platelet count < 90 × 109/L; • Serum total bilirubin >1.5×upper limit of normal (ULN); total bilirubin >3×ULN in case of determined Gilbert syndrome (unconjugated hyperbilirubinemia) or hepatic metastasis; • Serum alanine aminotransferase (ALT) and/or aspartate aminotransferase (AST) >2.5×ULN (ALT and/or AST >5×ULN for those with liver metastasis); • Blood creatinine >1.5×ULN, or creatinine clearance < 50 mL/min (calculated according to Cockcroft-Gault formula, see Appendix 2 for details.); • International normalized ratio (INR) > 1.5 and activated partial thromboplastin time (APTT) >1.5 × ULN.
	8	Having cardiovascular disease meeting any one of the following conditions: • Mean of three QTcFs on electrocardiography (ECG) >470msec under rested state, as calculated using Fridericia formula, see Appendix 9 for the details; • Any clinically important abnormalities in rhythm, conduction or morphology of resting ECG, e.g., complete left bundle branch block, Ⅲ degree conduction block, Ⅱ degree conduction block, PR interval>250 msec, etc.; • Various factors that may increase the risk of prolonged QTc or arrhythmic event, for example, heart failure, hypokalemia, congenital long QT syndrome, sudden death for unknown reason in the first-degree relative before 40 years old in the family history; • Left ventricular ejection fraction (LVEF) < 50%;
	9	HBV, HCV or HIV active infection. All the subjects will be screened for HBV, HCV or HIV infection in screening period; • HBsAg (+) and HBV DNA ≥1000 cps/mL (or 200 IU/mL) or HBV DNA higher than lower limit of quantification (LLQ) (if LLQ of HBV DNA is >1000 cps/mL); • anti-HCV antibody positive and HCV RNA positive; • HIV antibody positive;
	10	Known previous history of interstitial lung disease, drug induced interstitial lung disease, radiation pneumonitis requiring steroid treatment; or acute attack of or ongoing pulmonary symptoms that are considered by investigators as unsuitable for enrollment, or high-risk factors that are judged as possible interstitial lung disease and unsuitable for enrollment at baseline.
	11	Known or suspected allergy to the study drug and/or its

		excipients.
	12	Female patients in pregnancy or lactation.
	13	Presence of any disease or medical condition that is unstable or may affect its safety or compliance with the study, as considered by investigators, for example, uncontrolled hypertension, uncontrolled diabetes, active bleeding diathesis.

Supplementary Figure 1. Kaplan-Meier curve for progression-free survival assessed by independent review center in the per-protocol set

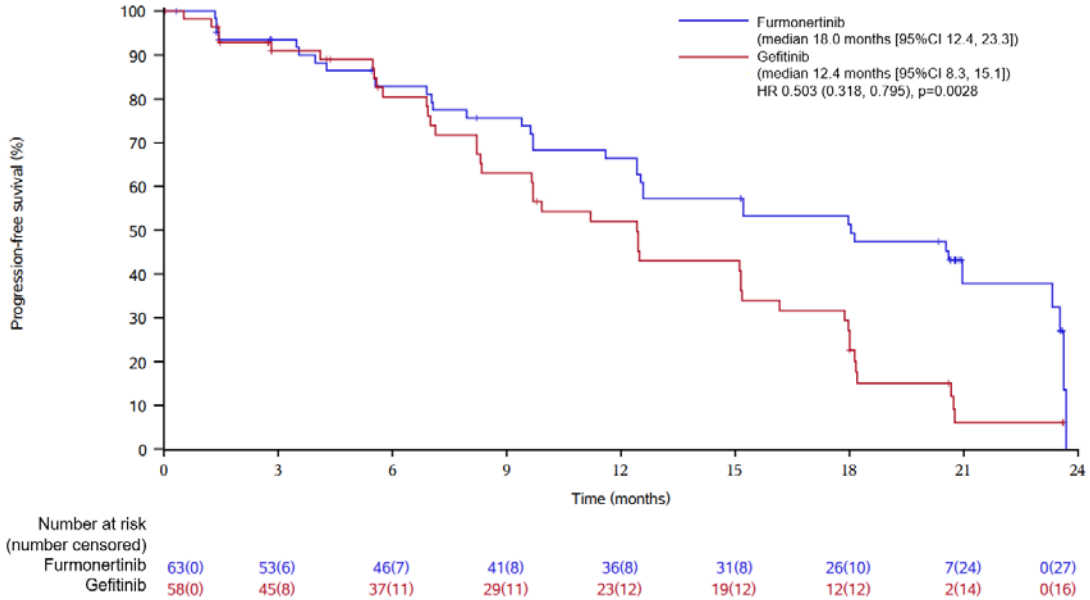


Supplementary Figure 2. Kaplan-Meier curve for progression-free survival assessed by investigator in the full analysis set

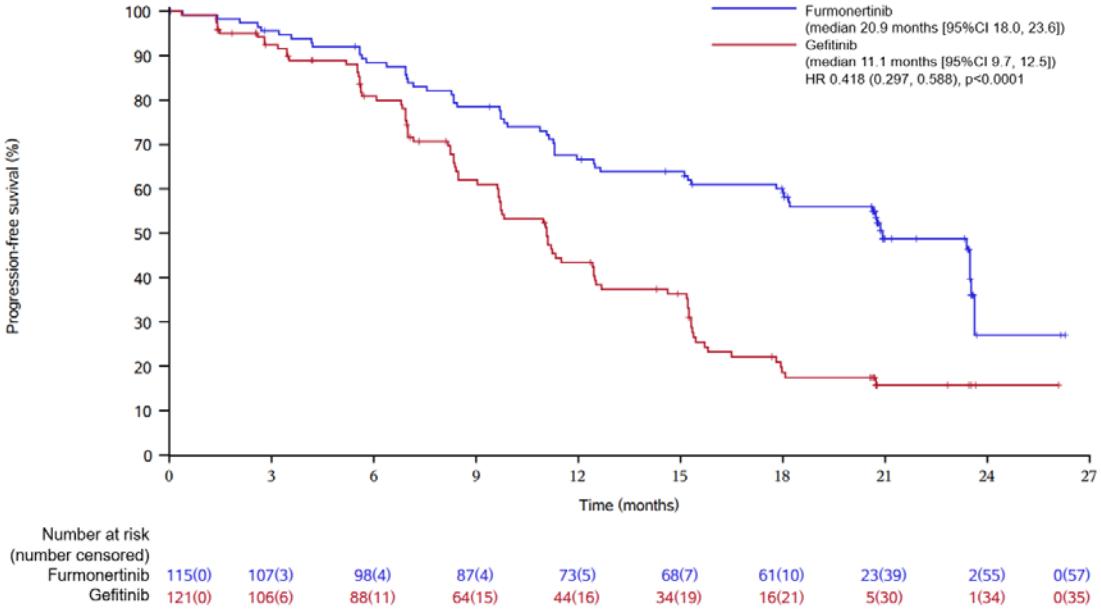


Supplementary Figure 3. Kaplan-Meier curve for progression-free survival in patients with and without central nervous system (CNS) metastases at baseline. CNS metastases was defined as CNS lesions judged by investigator as well as CNS histories irrespective of CNS lesions at baseline

A. Patients with CNS metastases

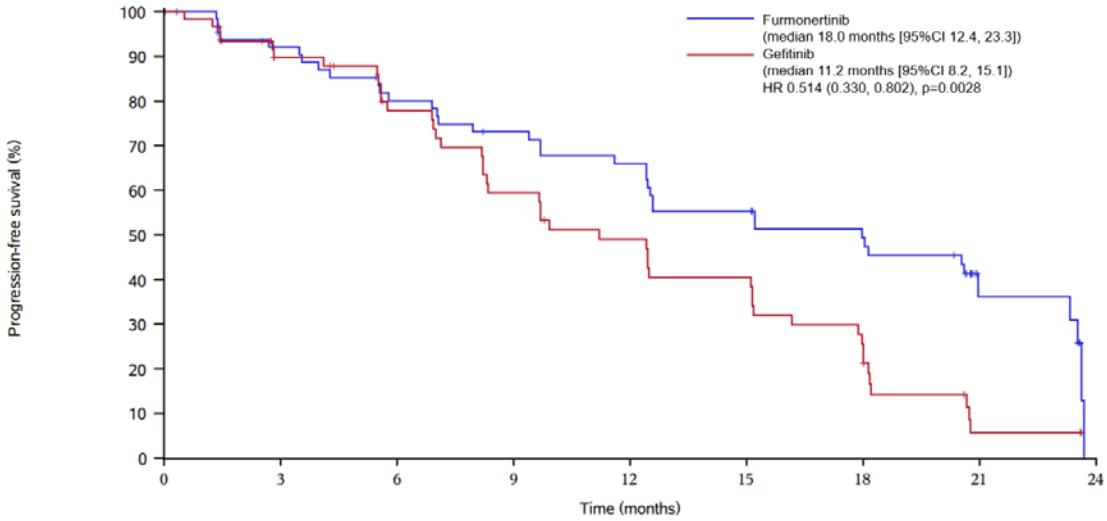


B. Patients without CNS metastases



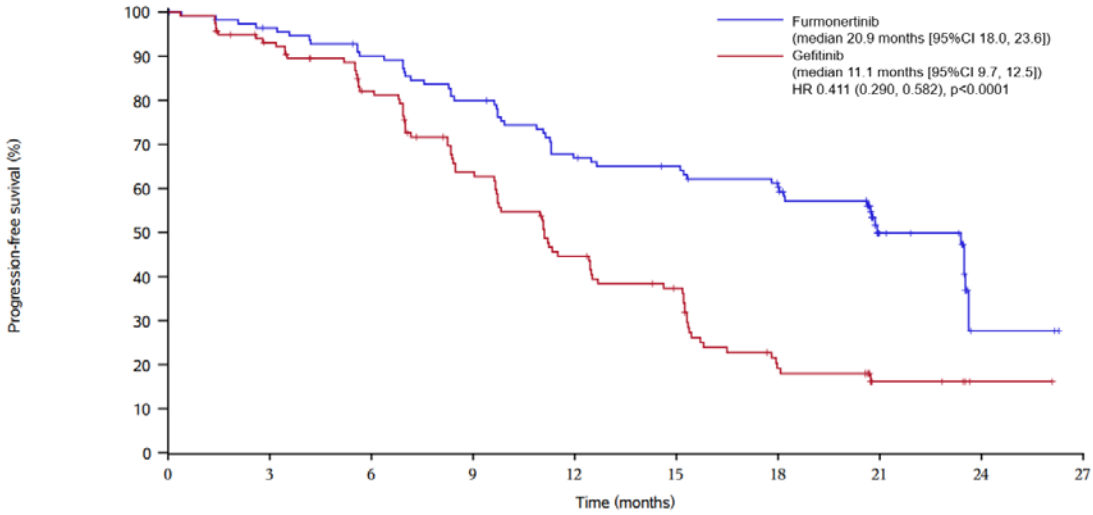
Supplementary Figure 4. Kaplan-Meier curve for progression-free survival in patients with and without central nervous system (CNS) metastases. CNS metastases was defined as measurable or non-measurable CNS lesions assessed by independent review center at baseline

A. Patients with CNS metastases



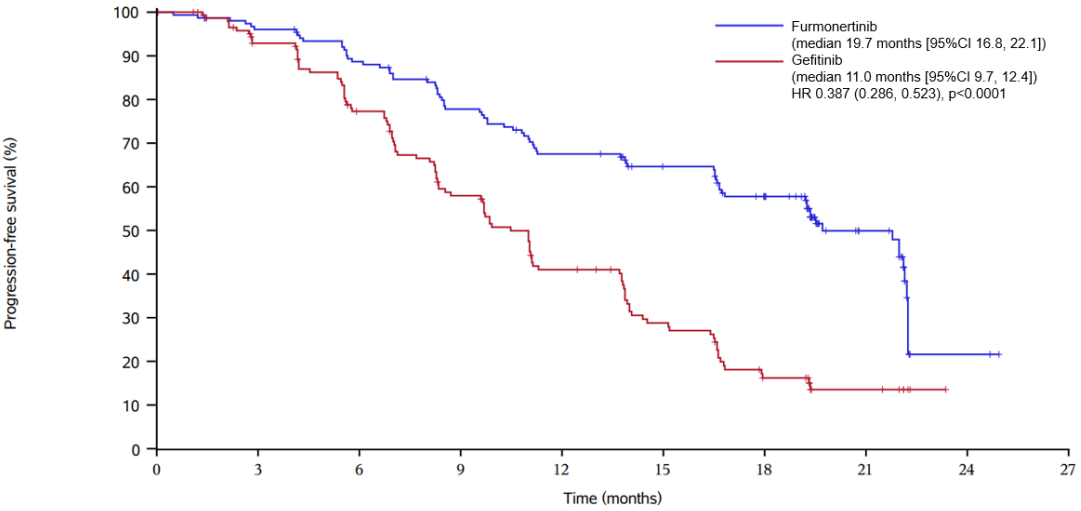
Number at risk (number censored)									
Furmonertinib	65(0)	54(6)	46(7)	41(8)	37(8)	31(8)	25(11)	7(24)	0(27)
Gefitinib	62(0)	47(9)	38(12)	29(12)	23(13)	19(13)	12(13)	2(15)	0(17)

B. Patients without CNS metastases



Number at risk (number censored)										
Furmonertinib	113(0)	106(3)	98(4)	87(4)	72(5)	68(7)	62(9)	23(39)	2(55)	0(57)
Gefitinib	117(0)	104(5)	87(10)	64(14)	44(15)	34(18)	16(20)	5(29)	1(33)	0(34)

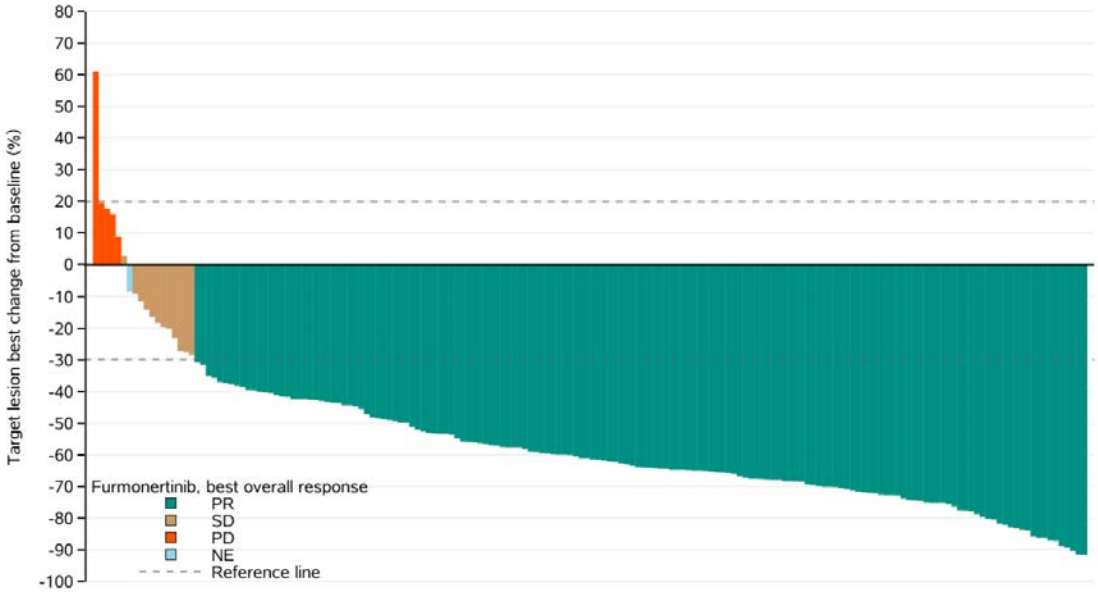
Supplementary Figure 5. Kaplan-Meier curve for duration of response assessed by independent review center



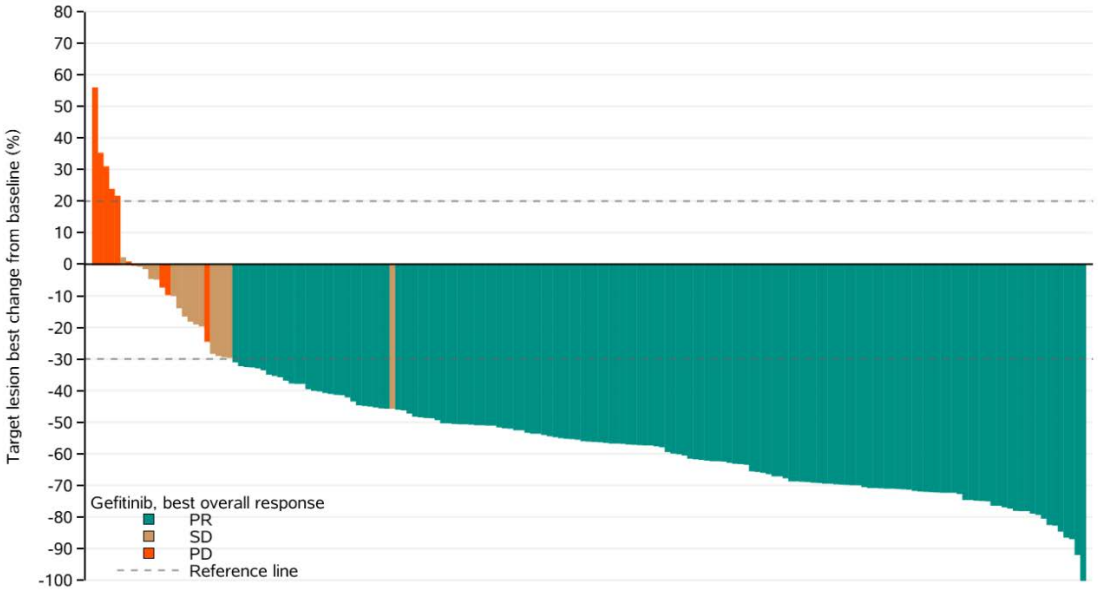
Number at risk (number censored)										
Furmonertinib	158(0)	145(7)	132(9)	114(11)	98(12)	86(20)	72(25)	26(64)	2(79)	0(81)
Gefitinib	151(0)	127(14)	101(19)	74(21)	50(24)	33(27)	16(30)	7(37)	0(44)	0(44)

Supplementary Figure 6. Waterfall plot for best percentage change in target lesion size in the full analysis set assessed by IRC

A. Furmonertinib group

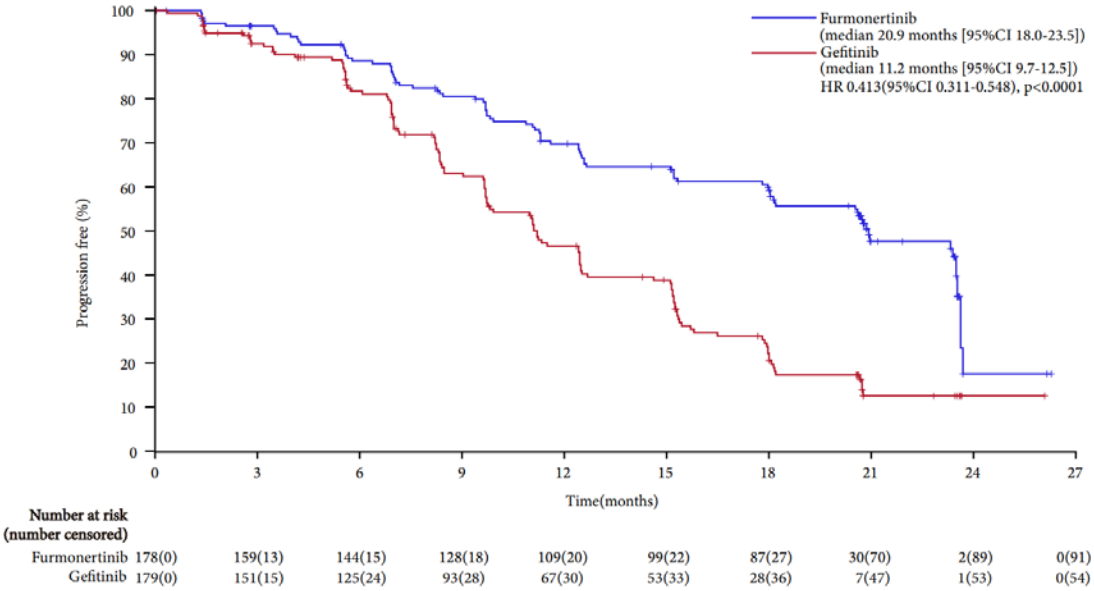


B. Gefitinib group



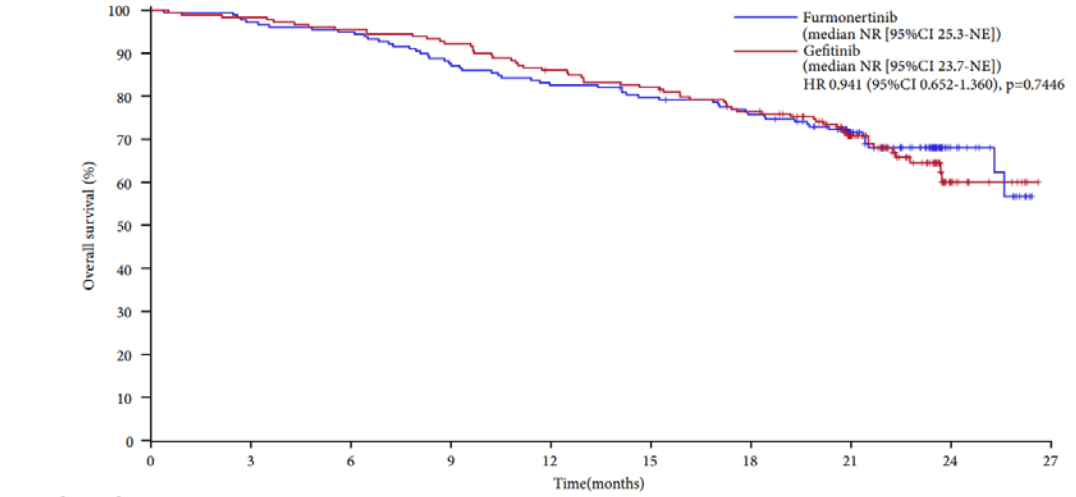
IRC=independent review center.

Supplementary Figure 7. Kaplan-Meier curve for time to progression assessed by IRC in the full analysis set



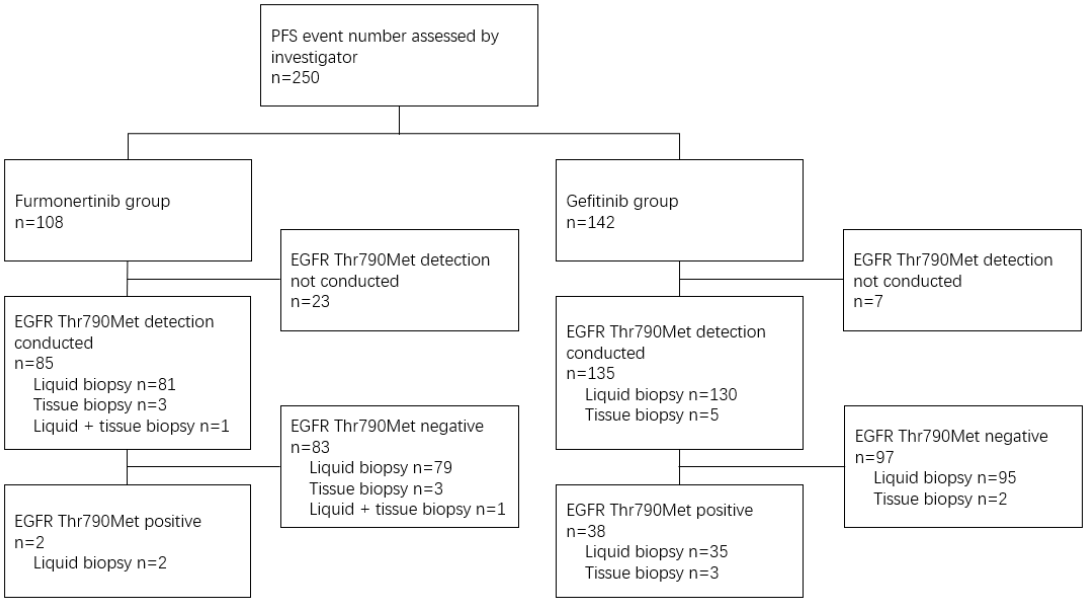
IRC= independent review center.

Supplementary Figure 8. Kaplan-Meier curve for overall survival



Number at risk (number censored)										
Furmonertinib	178(0)	173(0)	169(0)	156(0)	147(0)	141(1)	133(2)	87(41)	19(105)	0(122)
Gefitinib	179(0)	176(0)	171(0)	165(0)	153(1)	146(1)	134(3)	87(41)	16(104)	0(120)

Supplementary Figure 9. EGFR Thr790Met detection results post disease progression by central lab.



The EGFR Thr790Met status was detected with cobas *EGFR* Mutation Test (version 2, Roche Molecular Systems, Pleasanton, CA, USA).

Protocol

Please visit the following website for the full protocol:
https://www.allist.com.cn/upload/202204/25/202204251339503946.pdf